🔍 Minor Project

📌 **Task3:** Malware Detection tool
⏳ **Deadline:** 15th Feb, 11:59 PM

🌐 Malware:

Malware is software designed to harm, exploit, or compromise a system.

Example include:

- ➔ Viruses
- ➔ Worms
- ➔ Trojans
- ➔ Ransomware
- ➔ Spyware

🔑 Malware Detection Tool:

A malware detection tool scans files, applications, or systems to detect malicious threats. It does this by:

- Checking **file signatures** against a known malware database.
- Monitoring **suspicious behavior** (e.g., unauthorized access).
- Detecting **anomalies in system activity**.

🔑 Hash Signatures

A **hash signature** is a unique, fixed-length code generated using a **cryptographic hashing algorithm** (such as SHA-256) from a file's contents. It acts like a **fingerprint** for a file.

Why SHA-256?

- **Collision-resistant**

- **Highly secure (256-bit encryption)**
- **Widely trusted**
- **Ideal for cryptographic security and malware detection**

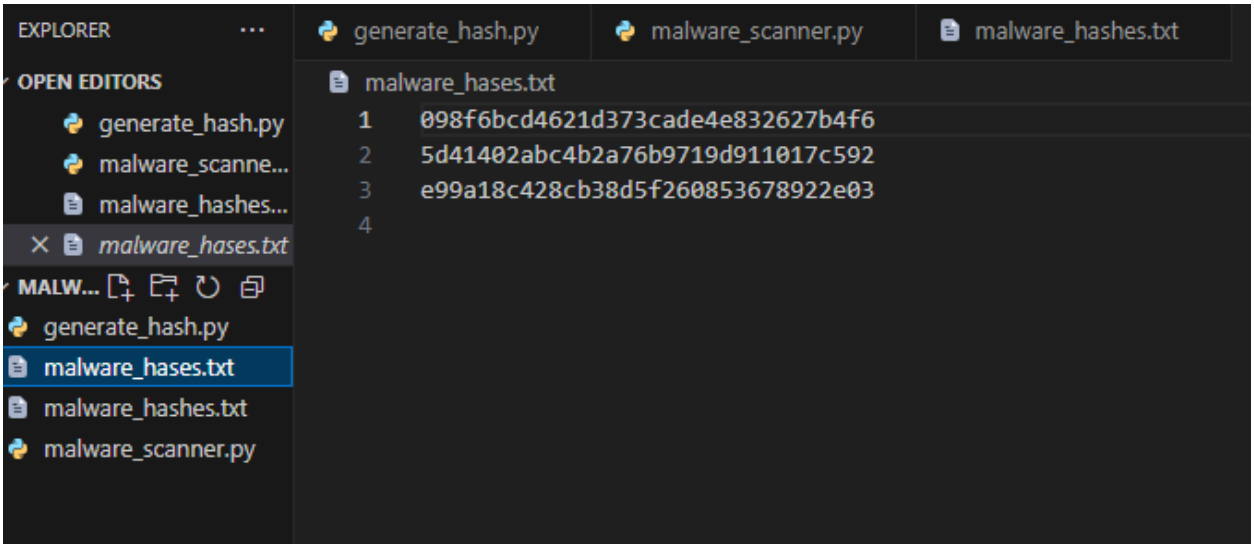
Project structure:

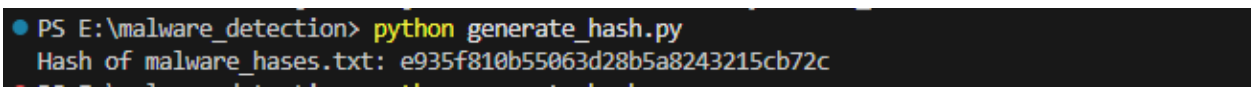
malware_detection

```
├── generate_hash.py       # Script to generate file hashes
├── malware_hashes.txt     # List of known malware hashes
├── malware_scanner.py     # Malware detection scanner
├── README.md              # Documentation
```

Step 1: creating malware_hases.txt file

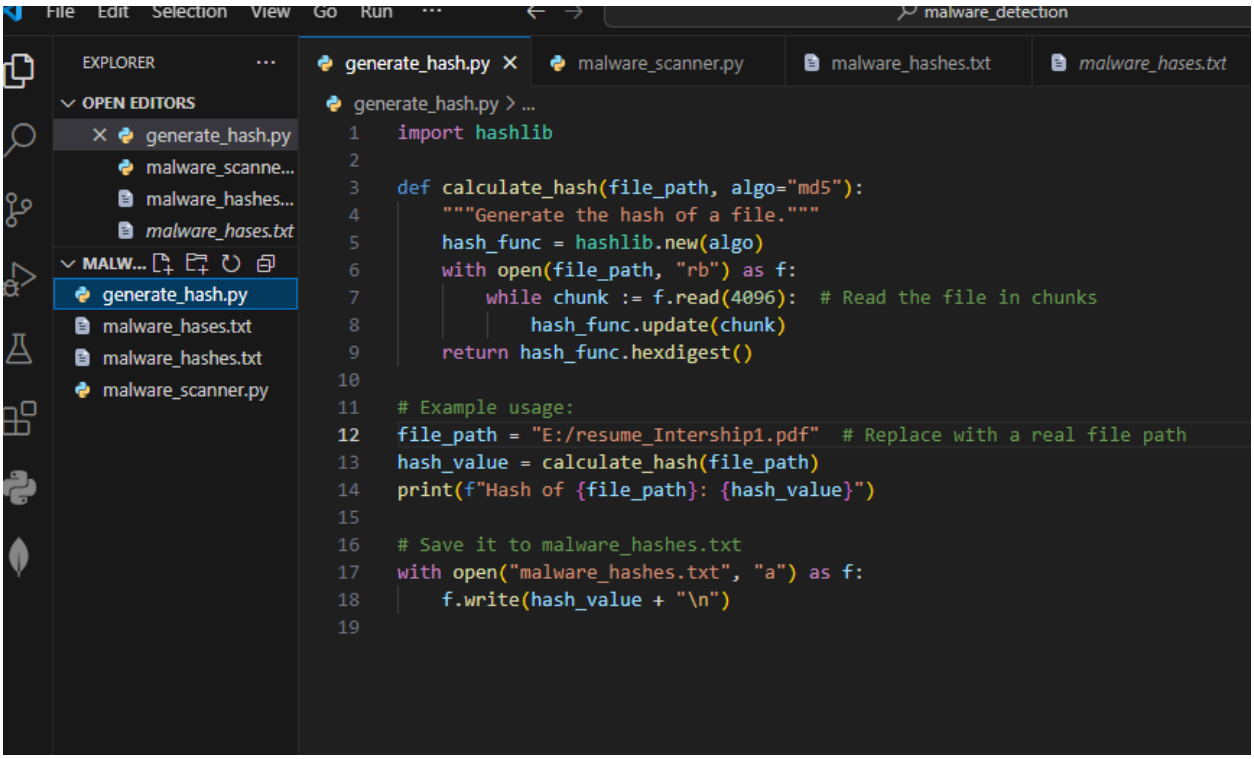
The first step is to create a file named `malware_hashes.txt`, which will store hash signatures of known malware files.





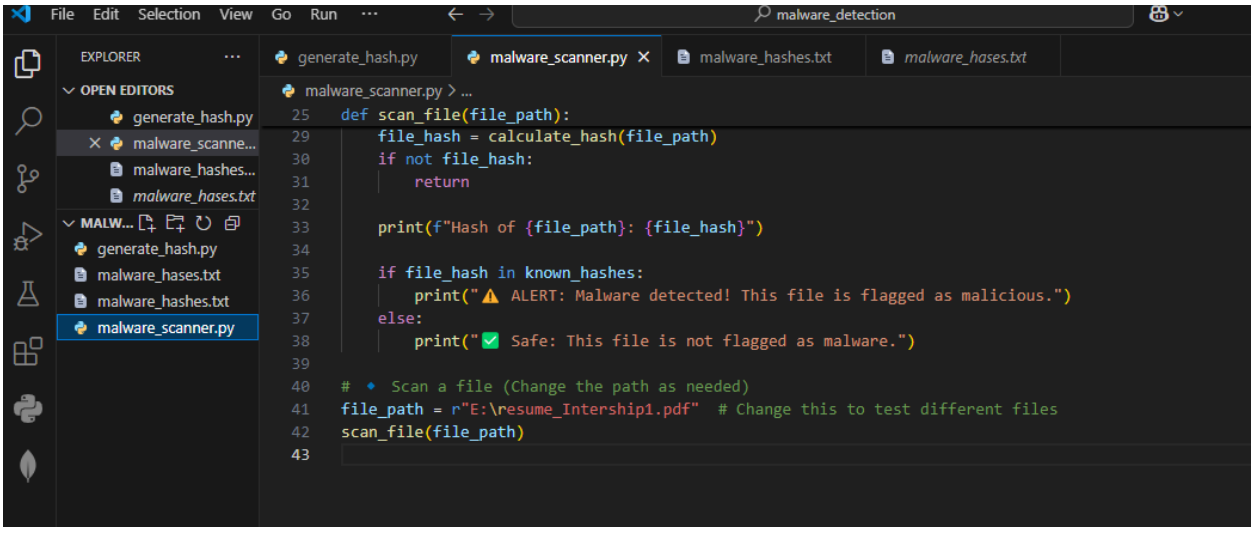
Step 2: Generating Hashes for Files:

We created a script `generate_hash.py` to generate SHA-256 hashes for any file



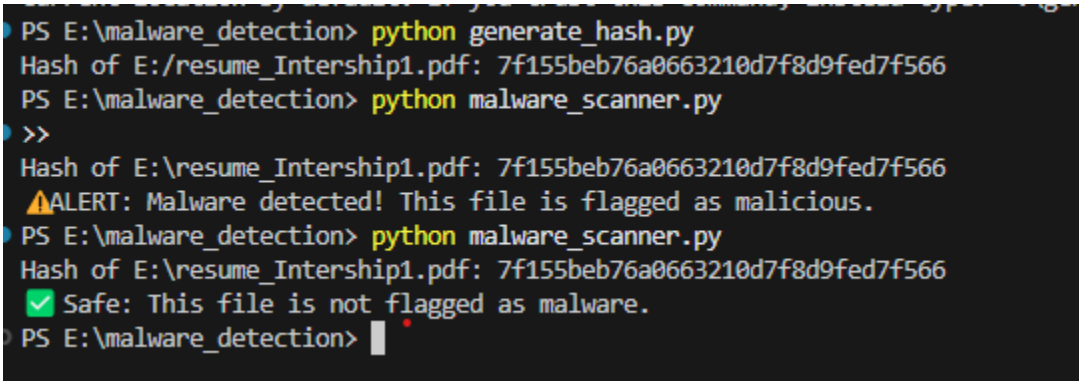
Step 3: Scanning for Malware:

then created `malware_scanner.py`, which compares the hash of a file with those stored in `malware_hashes.txt`.



Step 4: Running the Malware Scanner

- Run `generate_hash.py` to get the hash of a file.
- Run `malware_scanner.py` to check if the file is malware.



☑ Observations and Recommendations

- The scanned system has common web services running (HTTP, HTTPS).
- No immediate vulnerabilities detected, but regular security patches are recommended.
- Secure SSH by restricting access and using key-based authentication.
- Firewall rules should be reviewed to ensure no unnecessary open ports.

⌨ Conclusion

his project successfully implements a **basic malware detection tool** using hash signatures. The scanner helps identify malware by comparing file hashes with a known database of malware hashes.

📝 **Submitted By:** [Deepika Deshmukh]
📅 **Date:** 16th Feb
📂 **Submission: WEE4 Day 3**